

Comprehensive Analysis of Fruit Detection, Ripeness Assessment, and Mass Estimation using YOLOv7, ResNet, and ViT Models

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Abstract—China became the world’s largest apple producer, accounting for over 50% of the global planting area and output. The increasing urbanization has led to a decline in agricultural workers and higher labor costs, prompting the inevitable trend of employing picking robots for apple harvesting, supported by the widespread use of computer vision technology in agricultural automation. This Paper employs a comprehensive approach using YOLOv7, ResNet, and ViT models for fruit detection, ripeness assessment, and quality estimation. Initially, the YOLOv7 model processes input images to generate detection boxes identifying apples and further quantifies the number of apples in each image. Subsequently, based on the detection boxes from the YOLOv7 model, color analysis is performed to calculate fruit ripeness, presenting the results in the form of a histogram for a more intuitive understanding of ripeness distribution. Next, by calculating the red pixel area of identified apples, the study estimates fruit quality and presents the results as a histogram, illustrating the distribution of fruits of varying quality. Finally, the ResNet and ViT models are used for image classification, comparing their accuracy on a validation set to evaluate their performance differences in fruit image classification tasks. This integrated process provides a comprehensive and accurate analysis for fruit detection and assessment, offering robust support for subsequent research in fruit quality analysis and classification.

Keywords—Computer vision, YOLOv7, ViT, Detection boxes

I. INTRODUCTION

After years of rapid development, China became the world’s largest apple producer in 2019, with apple planting area and output accounting for more than 50% of the world’s total. Currently, apple picking in China is still mainly done manually. However, with the rapid development of urbanization in China, the number of agricultural workers has decreased sharply, and labor costs have increased significantly, which has brought adverse effects on the rapid development of the apple industry. Therefore, the application

of picking robots to achieve the automation of apple picking is an inevitable trend. In recent years, computer vision technology has been widely used in agricultural automation production. The visual system is an important component of the apple-picking robot. Using the visual system to identify and locate the target apple is one of the key links for the robot to perform the picking task. To achieve efficient apple picking by picking robots, it is necessary to accurately and quickly identify and locate the target apple.

Numerous scholars have been working on computer vision. Humans and animals acquire most of their information through the visual system. Vision in machines is provided by computer vision technology. Today, vision-based technologies play a vital role in the automation of various processes. With improvements in computation, capabilities and algorithms, computer vision performs different tasks such as object detection, human and animal action recognition and image classification. The application of computer vision in various industries is evolving [1]. Ji et al [2] developed a computer vision system that automates material counting activities without applying any markings to the material. The type of material of interest was sheet metal, which has a simple, large rectangular shape but is difficult to detect. The use of computer vision technology can reduce costs due to the loss of high-value materials, eliminate repetitive work requirements for skilled labor, and reduce human error. A computer vision system is proposed and tested for the sheet metal picking process using a single camera on multiple stacks of sheet metal in a storage area. The results show that the proposed computer vision system counts the sheet metal picking pieces in real situations with an accuracy of 97.83% and a recall rate of 100%. Liu et al [3] reviewed the research progress of computer vision technology in abnormal fish detection in the last two decades. For the first time, the fish anomaly detection task was categorized into three external manifestations: fish physiological activity anomalies, trajectory anomalies, and surface feature anomalies. The application methods of traditional and deep learning methods in computer vision technology in these three research types are further summarized, and the classical algorithmic models commonly used in fish anomaly detection are comprehensively introduced. In addition, this paper summarizes several common methods and model accuracy evaluation indexes for

acquiring public datasets in aquaculture and highlights two methods used by researchers to collect field experimental data. Finally, based on the above work, this paper analyzes several challenges in fish anomaly detection, proposes feasible strategies for each challenge, and points out the importance of improving the model to effectively integrate and analyze data from multiple platforms. Vallés-Pérez et al [4] proposed a new nonmonotonic activation function: the modulus. Most research on nonlinearity has focused on monotonic functions. It is empirically demonstrated how by using the modal activation function on a computer vision task, the model generalizes better than other nonlinear models - with a 15% improvement in CIFAR100 accuracy and a 4% improvement in CIFAR10 accuracy compared to the best benchmark activation tested. Using the proposed activation function, the problem of vanishing gradients and dying neurons disappears because the derivatives of the activation function are always 1 or -1. The simplicity of the proposed function and its derivatives makes the solution particularly suitable for TinyML and hardware applications.

Liu et al [5] proposed a driver fatigue detection algorithm based on dynamic tracking of facial eyes and yawning using YOLOv7 called FEY-YOLOv7. A coordinate attention module was inserted into YOLOv7 to enhance its dynamic tracking accuracy by focusing on coordinate information. In addition, a small target detection head is added to the network architecture to enhance the feature extraction capability of small face targets such as eyes and mouth. In terms of computation, the YOLOv7 network architecture is greatly simplified to achieve high detection speed. Using the proposed PERYAWN algorithm, driver states are labeled and detected by four categories: eyes open, eyes closed, mouth open, and mouth shut. In addition, a bootstrap image filtering algorithm is used to enhance the image details. The proposed FEY-YOLOv7 is trained and validated on an RGB infrared dataset. The results show that FEY-YOLOv7 has a mAP of 0.983 and an FPS of 101. Chen et al [6] proposed a recognition method based on the improved YOLOv7 model for detecting inflorescences at the apple bud, primordial and full bloom stages. In order to reduce the parameters, the Efficient Layer Aggregation Network (ELAN) in YOLOv7 was replaced by a residual network structure containing three convolutional layers. Squeeze excitation network (SENet) and coordinate attention (CA) were embedded in the last layer of the backbone network and the head network, respectively, to improve the recognition accuracy and sensitivity of apple inflorescences. Calculate the distance between the prediction frame and the true value frame more accurately. Refine the regression inference bias using the SIOU bounding box regression loss function to improve the bounding box prediction accuracy. In the detection head network, 80×80 detection heads were added to improve the recognition of small-scale apple inflorescences. Finally, the apple inflorescence phenology dataset was established. The results of the ablation experiments show that appropriate techniques can bring additional performance enhancement to the model. The proposed model outperforms the three models proposed in previous studies (YOLOv5s, improved YOLOv5 and YOLOv7). It achieved the best performance with 80.1% mAP and 42.58 frames per second (fps) recognition speed. The practicality and robustness of the recognition method were verified by developing a system for the detection and recognition of phenological apple inflorescences. Liu et al [7] proposed a method for the

detection of oil tea trunks based on an improved YOLOv7 network. First, we integrated the attention mechanism in the trunk and head layers of YOLOv7 to enhance the feature extraction of the relay so that the network can focus on the relevant target objects. Second, a Facol-EIoU-based weighted confidence loss function is designed to replace the original loss function in the improved YOLOv7 network; the modification aims to improve the detection performance of oil tea trunks.

In this paper, YOLOv7, ResNet and ViT models are combined for fruit detection, maturity assessment and quality estimation. First, the images are processed using the YOLOv7 model to generate detection frames to identify apples and count the number of apples in each image, which in turn plots the distribution histogram of the number of apples. Then, the YOLOv7 model is used to generate a detection frame for each identified apple, calculate the fruit ripeness by color analysis, and present the ripeness results in the form of a histogram. Next, the area of the red pixels of the recognized apples is calculated and the quality of the apples is estimated and the histogram of the apple quality distribution is plotted. Finally, the ResNet model and the ViT model are used for image classification and their accuracy on the validation set is compared. In this paper, with <https://publicqn.saikr.com/2023/11/23/contest/54897b0ddcea b12f261fe648854fc0421700689989602.zip?attnname=2023%20APMCM%20image data from 20Problems.zip> is analyzed as an example.

II. MODEL BUILDING

A. YOLOv7 modeling

Object detection is a fundamental task in computer vision that involves identifying and locating objects within an image or a video. Over the years, numerous algorithms and models have been developed to address this challenge, each aiming to improve accuracy, speed, and robustness. One such notable advancement is the YOLO (You Only Look Once) series, which has gained popularity for its real-time object detection capabilities.

The YOLO series has seen several iterations, each building upon the strengths of its predecessors. YOLOv7, the focus of this discussion, represents a significant milestone in the evolution of the YOLO family. YOLOv7 integrates innovative techniques and architectural enhancements to further improve object detection performance.

At the core of YOLOv7 lies its architecture, which is designed to efficiently process images and accurately detect objects within them. The architecture of YOLOv7 is a culmination of various components, including backbone networks, feature pyramid networks, and detection heads, all working in harmony to achieve precise object localization and classification.

Backbone networks in YOLOv7 are responsible for extracting high-level features from the input images. These networks typically consist of convolutional layers, which progressively analyze the input data to capture intricate patterns and details. Notable examples of backbone networks used in YOLOv7 include Darknet and CSPDarknet, known for their effectiveness in feature extraction.

Feature pyramid networks play a crucial role in multi-scale object detection. They enable YOLOv7 to detect objects of varying sizes within an image by creating feature

pyramids that capture information at different scales. This is achieved through lateral connections and additional convolutional layers, allowing the model to effectively handle objects of different sizes and aspect ratios.

The detection heads in YOLOv7 are responsible for predicting bounding boxes, object classes, and confidence scores. These heads leverage the features extracted by the backbone and feature pyramid networks to make accurate predictions about the presence and location of objects within an image. YOLOv7 utilizes anchor boxes and advanced regression techniques to refine the predicted bounding boxes, leading to improved localization accuracy.

The success of YOLOv7 can be attributed, in part, to the sophisticated formulas and computations that underpin its object detection capabilities. One of the key formulas used in YOLOv7 is the calculation of the bounding box coordinates. This formula involves predicting the coordinates of the bounding boxes relative to anchor boxes and applying appropriate transformations to obtain the final bounding box predictions.

Let's denote the coordinates of an anchor box as (x_a, y_a, w_a, h_a) , where (x_a, y_a) represents the center of the anchor box, and (w_a, h_a) denote its width and height. The predicted offsets for the bounding box, denoted as $(\Delta x, \Delta y, \Delta w, \Delta h)$, are then used to adjust the dimensions and position of the anchor box to obtain the final predicted bounding box coordinates.

The formulas for predicting the coordinates of the bounding box are as follows:

$$x = \sigma(\Delta x) + x_a \quad (1)$$

$$y = \sigma(\Delta y) + y_a \quad (2)$$

$$w = e^{\Delta w} \times w_a \quad (3)$$

$$h = e^{\Delta h} \times h_a \quad (4)$$

In these formulas, σ represents the sigmoid function, which scales the offsets to the range $[0, 1]$, ensuring that the predicted coordinates lie within the bounds of the anchor box. The exponential function $e^{\Delta w}$ and $e^{\Delta h}$ are used to transform the predicted width and height offsets, allowing the model to accurately adjust the dimensions of the bounding box relative to the anchor box.

The derivation of the bounding box prediction formula involves a series of transformations and computations aimed at refining the initial anchor box predictions. At the heart of the derivation is the idea of predicting adjustments, or offsets, to the dimensions and position of the anchor box to obtain the final bounding box coordinates.

The derivation begins with the prediction of offsets $(\Delta x, \Delta y, \Delta w, \Delta h)$, which represent the adjustments needed to transform the anchor box into the actual bounding box containing the detected object. These offsets are typically obtained through regression techniques, where the model learns to predict the necessary adjustments based on the features extracted from the input image.

Once the offsets are predicted, they are applied to the coordinates and dimensions of the anchor box to obtain the final bounding box predictions. The sigmoid function σ is used to scale the predicted offsets $(\Delta x, \Delta y)$ to the range $[0, 1]$,

effectively adjusting the position of the bounding box within the anchor box.

Similarly, the exponential function is applied to the predicted offsets $(\Delta w, \Delta h)$, allowing the model to accurately adjust the width and height of the bounding box relative to the anchor box. This transformation ensures that the final bounding box dimensions are appropriately scaled based on the predicted adjustments.

B. ResNet modeling

Residual Networks, commonly known as ResNets, have revolutionized the field of deep learning by addressing the challenges of training very deep neural networks. Developed by Kaiming He et al., ResNets introduced a novel architecture that significantly improved the training and performance of deep convolutional neural networks (CNNs). This essay delves into the intricacies of ResNets, exploring their architecture, the concept of residual learning, and the mathematical formulations that underpin their success.

Training deep neural networks poses several challenges, including the vanishing gradient problem and the degradation problem. As networks grow deeper, it becomes increasingly difficult to optimize their training, leading to diminishing performance gains or even degradation (increased training error) despite additional layers. ResNets were introduced to mitigate these issues and enable the training of extremely deep networks, unlocking the potential for improved accuracy and feature representation.

At the core of ResNets lies the concept of residual learning, which fundamentally alters the way information flows through the network. Traditional neural network architectures learn underlying representations of the input data through a series of stacked layers. However, ResNets introduce residual connections, also known as skip connections, which allow the network to learn residual functions with respect to the layer inputs. This approach facilitates the training of much deeper networks by enabling the direct flow of information through the residual connections, thus mitigating the vanishing gradient problem.

The architecture of ResNet is characterized by its residual blocks, each of which consists of multiple convolutional layers and identity mappings. These residual blocks form the building blocks of the network and are instrumental in enabling the efficient training of very deep networks. The key components of the ResNet architecture are the residual connections and the skip connections, which play a pivotal role in reshaping the flow of information and enabling the successful training of deep networks.

The residual function, denoted as $\mathcal{F}$, captures the deviation between the input and the desired output within a residual block. This function represents the internal transformations applied to the input data, allowing the network to learn the residual mapping. Mathematically, the residual function can be expressed as:

$$\mathcal{F}(\text{Input}) = \text{Output} - \text{Input} \quad (1)$$

where *Input* represents the input to the residual block, and *Output* denotes the output of the block. By learning the residual function, the network effectively captures the difference between the input and the desired output, enabling the training of deeper networks with improved performance.

The skip connection, which directly transmits the input to a later layer, can be mathematically represented as an identity mapping. This formulation ensures that the input information bypasses intermediate layers, preserving the gradient flow during training. The skip connection can be expressed as:

$$\text{Output} = \mathcal{F}(\text{Input}) + \text{Input} \quad (2)$$

where $\mathcal{F}(\text{Input})$ represents the transformation applied within the residual block, and Input denotes the input to the block. By incorporating skip connections, ResNets maintain the flow of information and gradients, enabling the successful training of very deep networks.

C. ViT modeling

The Vision Transformer (ViT) represents a groundbreaking approach to computer vision tasks by leveraging the power of transformer architectures, originally popularized in natural language processing. Developed by Dosovitskiy et al., ViT has demonstrated remarkable performance in image classification, object detection, and other vision-related tasks. This essay delves into the intricacies of ViT, exploring its architecture, the concept of self-attention, and the mathematical formulations that underpin its success.

The Transformer architecture, initially designed for sequence modeling in natural language processing, has been successfully adapted to address computer vision tasks through ViT. Unlike traditional convolutional neural networks (CNNs), which rely on convolutional and pooling layers to process visual information, ViT introduces a novel paradigm by treating images as sequences of patches, enabling the application of self-attention mechanisms to capture global dependencies within the input image.

The architecture of ViT consists of a stack of transformer blocks, each of which comprises multiple layers of self-attention and feedforward neural networks. The key innovation lies in the initial processing of the input image, where it is divided into fixed-size patches, which are then linearly embedded to generate token sequences. These token sequences are subsequently fed into the transformer encoder, enabling the model to capture relationships and dependencies across the entire image.

The input image is divided into non-overlapping patches, each of which is linearly embedded to produce token embeddings. Mathematically, the patch embeddings can be represented as follows:

Let $X \in \mathbb{R}^{H \times W \times C}$ denote the input image, where H represents the height, W represents the width, and C represents the number of channels. The image X is divided into non-overlapping patches, resulting in a sequence of 2D embeddings $x_i \in \mathbb{R}^E$, where E represents the dimension of the patch embeddings.

The self-attention mechanism forms the crux of the transformer architecture and plays a pivotal role in capturing global dependencies within the input sequences. Within ViT, self-attention enables the model to learn contextual relationships between different patches, facilitating the extraction of meaningful visual features.

Mathematically, the self-attention mechanism computes the attention weights and generates context-aware representations of the input tokens. Given a sequence of

token embeddings $x_1, x_2, \dots, x_N$, the self-attention mechanism computes the attention weights $A \in \mathbb{R}^{N \times N}$ and generates the context-aware representations $Z \in \mathbb{R}^{N \times E}$, where N represents the number of tokens and E denotes the dimension of the embeddings.

The patch embeddings can be mathematically represented as:

$$x_i = \text{Emb}(X_{\text{patch}_i}) \quad (1)$$

where X_{patch_i} represents the i -th image patch and $\text{Emb}()$ denotes the linear embedding function.

The self-attention mechanism computes the attention weights A and generates the context-aware representations Z as follows:

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V \quad (2)$$

$$Z = A \cdot X \quad (3)$$

where Q , K , and V represent the query, key, and value matrices, and d_k denotes the dimension of the key vectors.

The transformer encoder processes the input token sequences through self-attention and feedforward networks, enabling the extraction of high-level features:

$$\text{Encoder}(X) = \text{FFN}(\text{LayerNorm}(\text{SelfAttention}(X)) + X) \quad (4)$$

where $\text{SelfAttention}()$ represents the self-attention mechanism, $\text{FFN}()$ denotes the feedforward neural network, and $\text{LayerNorm}()$ signifies the layer normalization operation.

III. ANALYSIS

A. Quantitative identification

Using the yolov7 model, all images are passed into this model and each identified apple is boxed out to get the image containing the detection box. The recognition effect is shown in Figure 1.



Fig. 1. Identify the detection block diagram corresponding to each apple

From the yolov7 model, the number of detection frames in each image is generated, and each identified apple corresponds to one detection frame respectively, and the number of detection frames is the number of identified apples.

After the detection of 200 pictures, the data were analyzed using SPSS software to obtain a mean value of 10.39 for the number of apples, a minimum value of 1, and a maximum value of 105, with the number of apples as the

horizontal coordinate, and the number of pictures plotted as the vertical coordinate in a histogram as shown in Figure 2.

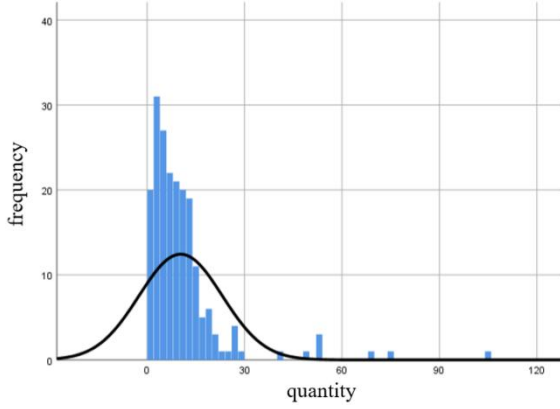


Fig. 2. Histogram of apple count distribution

The number of each image is the horizontal coordinate and the number of apples detected in each image is the vertical coordinate, a histogram of the distribution of the number of apples is shown in Figure 3.

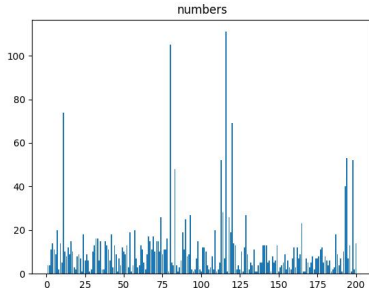


Fig. 3. Histogram of apple count distribution by using

B. Maturity Identification

We divided the maturity level into three parts at the boundaries of 0.2 and 0.25, corresponding to unripe, semi-ripe and ripe. The image is converted to HSV color space and corresponding color boundaries are created based on predefined color ranges. The color boundaries are then thresholded to obtain a mask image of the corresponding color. The number of pixels of each color was calculated by performing pixel counting on the mask image and the maturity level was calculated based on the number of red, green and yellow pixels and the weight values. As shown in Figure 4.

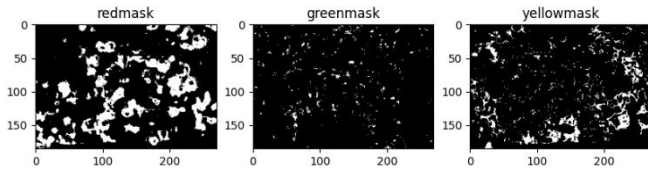


Fig. 4. Apple ripening process chart

We can draw a histogram of the maturity distribution of apples, with the horizontal coordinate in Figure 5 being the serial number of each image and the vertical coordinate being the maturity of the detected apples in each image, as generated by the apple maturity code.

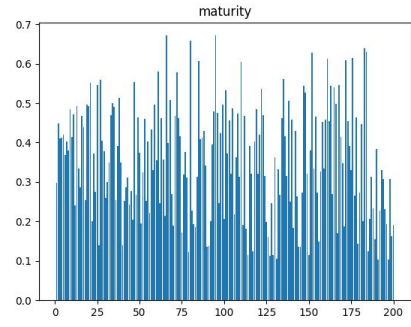


Fig. 5. Histogram of maturity distribution of apples

C. Quality Recognition

We will use a 25-cent dollar coin as a reference and make sure that the coordinates of his associated border can be simply measured to get a conversion relation between mass and pixels, and then get the mass distribution from the apple's pixel distribution and make an histogram based on this conversion relation. This coin is 0.955 inches and has a mass of 5.67 g. Now suppose our (in pixels) is calculated to be 150 pixels wide (based on its associated border). Then by, $P = \frac{W_o}{W_k}$ and $P = 150px/0.955in = 157px$ and $P_g = 5.67g/0.955in \approx 5.937px/g$, so there are about 157 pixels per 1ft in our image, with a mass of about 5.937g Using this ratio, we can calculate the size and mass of the object in the image.

Draw a histogram of the distribution of the mass of the identified apples in each picture, with the horizontal coordinate being the serial number of each picture and the vertical coordinate being the calculated mass of the identified apples in each picture. As shown in Figure 6.

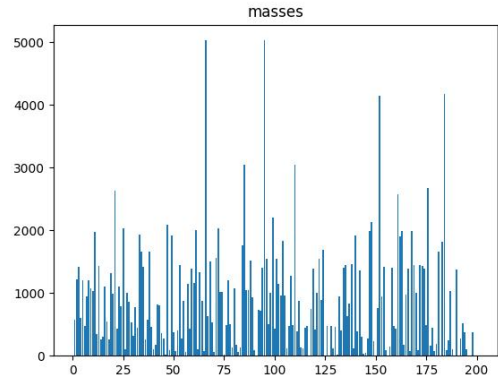


Fig. 6. Histogram of apple quality distribution

D. Comparison of identification accuracy

By comparing the accuracy of the resnet model and the vif model on the validation set we can get Figure 7 and Figure 8.

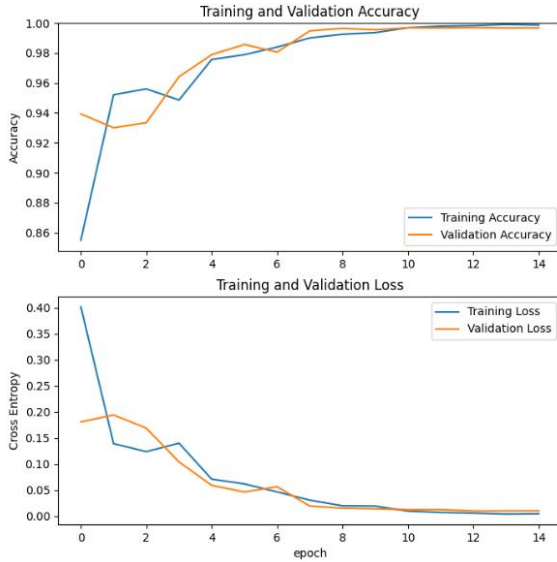


Fig. 7. Comparison of resnet model and vif model results

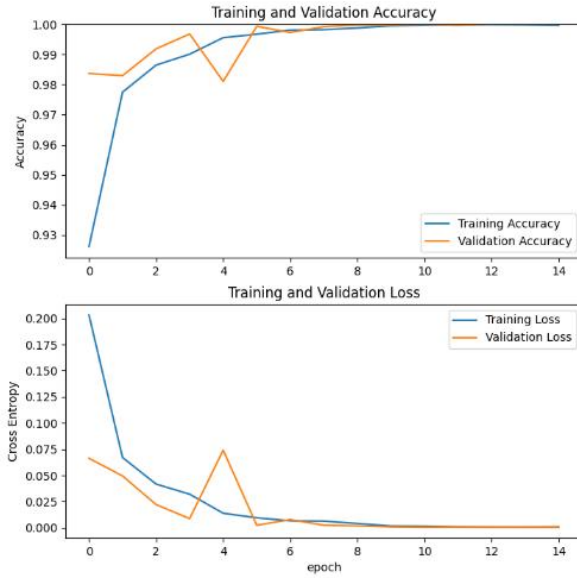


Fig. 8. Comparison of resnet model and vif model results

IV. CONCLUSION

In this study, a combination of YOLOv7, ResNet and ViT models was used for fruit detection, ripeness assessment and quality estimation. First, the input images were processed using the YOLOv7 model to generate detection frames to identify apples and further count the number of apples in each image. Subsequently, based on the detection frames generated by the YOLOv7 model, color analysis is performed to calculate the ripeness of the fruits, and the results are presented in the form of histograms for a more intuitive understanding of the distribution of ripeness. Next, the quality of the apples is estimated by calculating the area of the red pixels of the recognized apples, and the estimated results are presented as a histogram of the apple quality distribution to show the distribution of fruits with different qualities. Finally, the ResNet model and the ViT model are used for image classification, and their accuracies on the validation set are comparatively analyzed in order to assess the difference in their performance in the fruit image classification task. The whole process combines the features of different models to provide a comprehensive and accurate analysis for fruit detection and evaluation, which provides strong support for subsequent research on fruit quality analysis and classification.

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